

Kidney Disease Prediction



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PRESENTATION OUTLINE

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Abstract

Kidney disease is a critical health issue that often goes undetected until it reaches advanced stages. Early diagnosis is essential for effective treatment and improved patient outcomes. This system leverages machine learning algorithms to predict the likelihood of kidney disease based on patient data, ensuring timely detection. By providing accurate predictions, the system aids healthcare professionals in making informed decisions and improving the quality of care for patients.

The model is trained using Machine Learning algorithms, including classification techniques, to accurately predict kidney disease. A set of features such as patient health parameters (e.g., blood pressure, creatinine levels, etc.) are defined and trained on a relevant medical dataset to enhance accuracy and reliability.

Introduction

The Kidney Disease Prediction System is a cutting-edge healthcare tool that leverages machine learning algorithms to identify the likelihood of kidney disease in patients. The system's goal is to provide early predictions based on patient health data, enabling timely medical intervention and better disease management through advanced data analysis and classification techniques.

Objectives

- **Early Detection & Diagnosis** – Identify kidney disease in its early stages to enable timely medical intervention.
- **High Prediction Accuracy** – Develop a machine learning model that provides reliable and accurate predictions.
- **Feature Analysis & Risk Factors Identification** – Analyze key biomarkers (e.g., blood pressure, creatinine levels, proteinuria) that contribute to kidney disease.

- **User-Friendly Interface** – Design a simple and accessible interface for healthcare professionals and patients.
- **Integration with Medical Records** – Ensure the system can integrate with existing healthcare databases for better data management.
- **Health Recommendations & Alerts** – Provide personalized health suggestions and alerts based on prediction results.

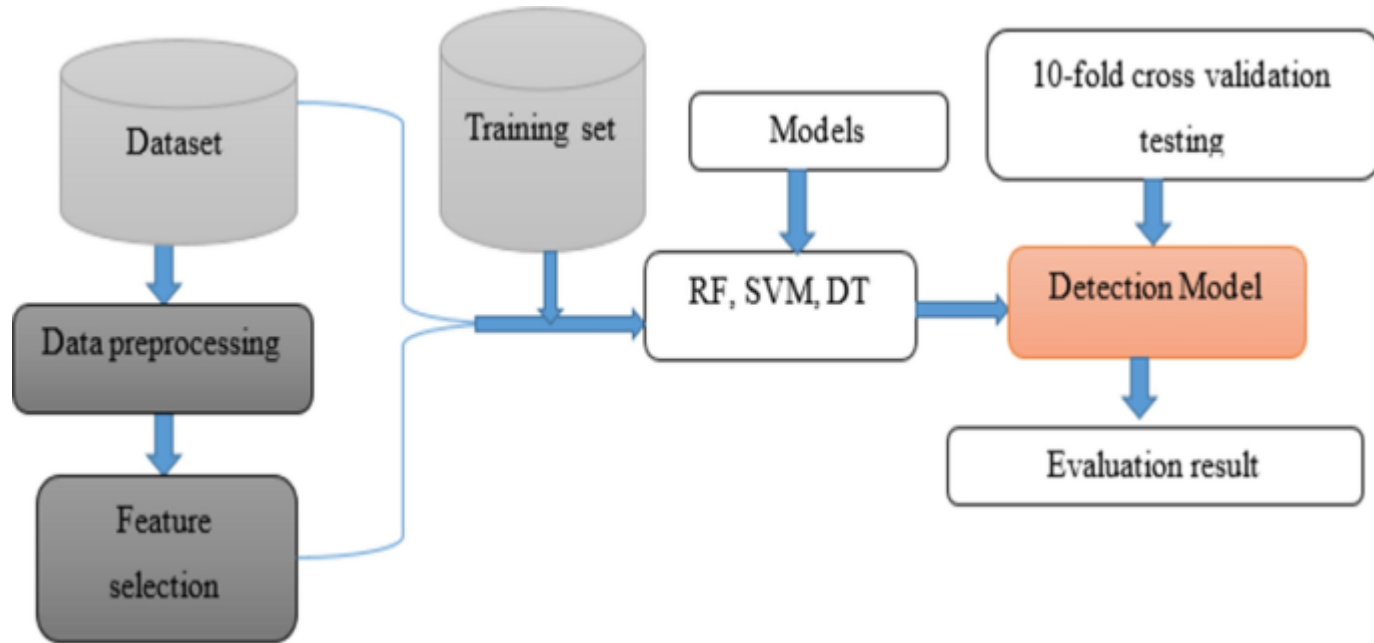
Proposed System

- Our system uses machine learning to predict kidney disease by analyzing medical data such as lab results and patient history.
- It improves on previous models by considering additional factors for more accurate predictions. The system also sends real-time alerts, raising awareness and helping patients take timely action.

Advantages:

- Recognizes traffic signs and provides a voice alert to the driver, warning of potential risks ahead.
- Enables the driver to take necessary actions or avoid dangerous locations.
- Helps prevent accidents by increasing driver awareness.
- Reduces driver distractions, allowing them to focus on the road.
- Enhances overall road safety by providing timely alerts.

System Architecture



Algorithm Implementation

1. Data Collection & Preprocessing

- Gather medical data (e.g., age, blood pressure, creatinine, glucose).
- Handle missing values and encode categorical data.
- Normalize numerical features for better performance.

2. Split Data into Training & Testing Sets

- Divide the dataset into 80% training and 20% testing.

3. Train the Logistic Regression Model

- Fit the model to training data to learn relationships between features and

kidney disease.

4. Make Predictions

- Use the trained model to predict whether a patient has CKD (1) or not (0).

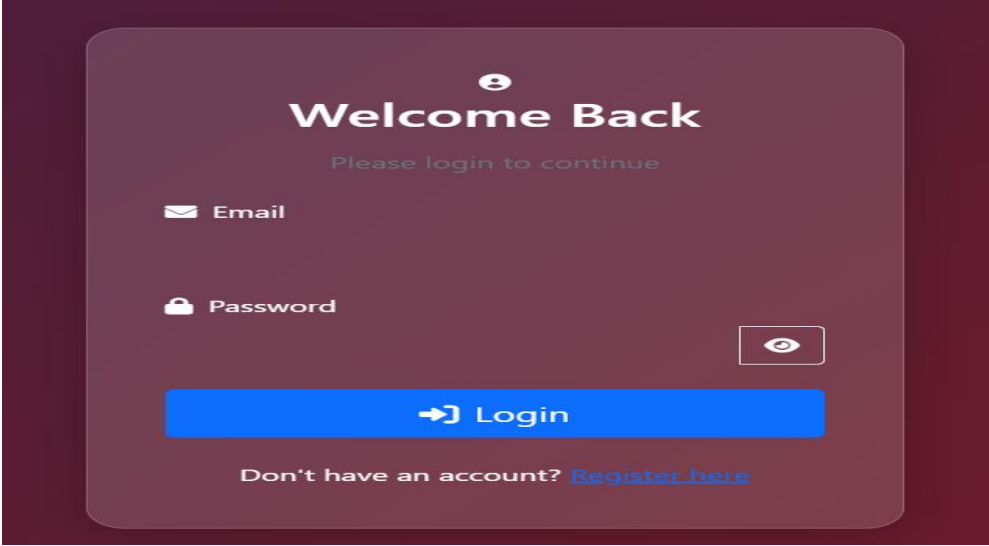
5. Evaluate Model Performance

- Measure accuracy, precision, recall, F1-score, and analyze the confusion matrix.

6. Deploy the Model

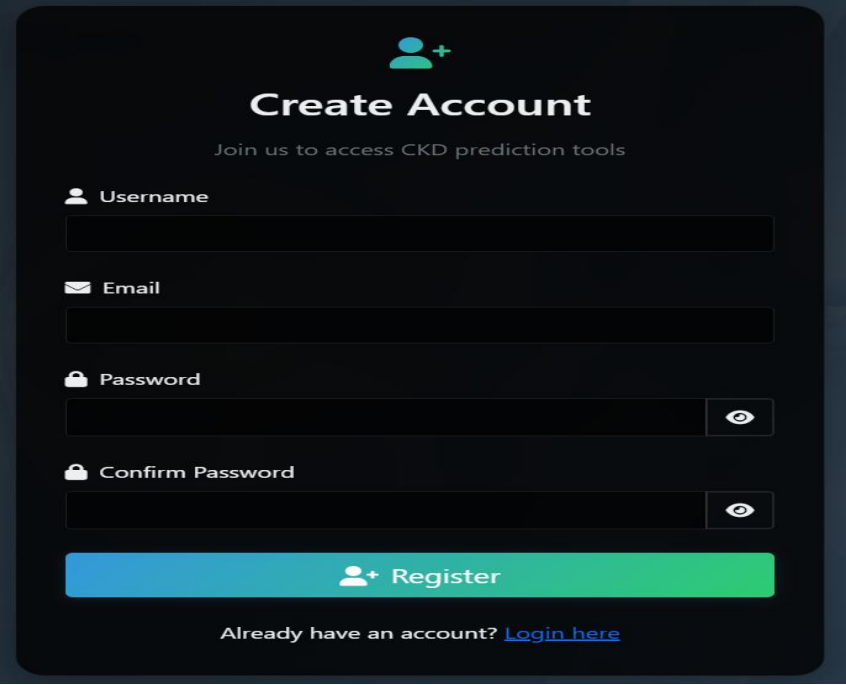
- Save the trained model and integrate it into healthcare applications for real-world use.

Experimental Results



A screenshot of a login page with a dark red background. In the center is a rounded rectangle with a lighter red background. At the top of this rectangle is a small user icon, followed by the text "Welcome Back" in bold. Below that is the text "Please login to continue" in a smaller, lighter font. There are two input fields: "Email" with an envelope icon and "Password" with a lock icon. To the right of the password field is a toggle icon (an eye inside a square). Below the input fields is a large blue button with a right-pointing arrow and the text "Login". At the bottom of the rectangle, it says "Don't have an account? [Register here](#)".

Figure 1 : Login Page.



The registration form is titled "Create Account" with a subtitle "Join us to access CKD prediction tools". It features four input fields: "Username", "Email", "Password", and "Confirm Password". Each field has a corresponding icon (person, envelope, and padlocks respectively). The "Password" and "Confirm Password" fields include toggle icons for visibility. A green "Register" button is positioned below the fields, and a link "Login here" is provided for existing users.

 **Create Account**
Join us to access CKD prediction tools

 Username

 Email

 Password
 

 Confirm Password
 

 Register

Already have an account? [Login here](#)

Figure 2 : Register Page.



Figure 3 : Home Page of Kidney Disease Prediction.

 **Upload Dataset**


Drag and drop your CSV file here
or


 **Dataset Requirements**

- ✓ File format: CSV only
- ✓ Maximum file size: 16MB
- ✓ Required columns:
 - age
 - blood_pressure
 - specific_gravity
 - albumin
 - blood_glucose
 - blood_urea
 - serum_creatinine
 - sodium
 - potassium
 - hemoglobin
 - target (0 for no CKD, 1 for CKD)

Figure 4 : Dataset Uploading and Training Page.

Kidney Disease Prediction Form

Model Accuracy: 100.0%

Age	Blood Pressure
Patient's age in years	Blood pressure in mm/Hg
Specific Gravity	Albumin
Urine specific gravity (1.005-1.025)	Albumin level (0-6)
Blood Glucose	Blood Urea
Blood glucose level in mg/dL	Blood urea level in mg/dL
Serum Creatinine	Sodium
Serum creatinine level in mg/dL	Sodium level in mEq/L
Potassium	Hemoglobin
Potassium level in mEq/L	Hemoglobin level in g/dL

Predict

Kidney Disease Prediction Form

Model Accuracy: 100.0%

Age	Blood Pressure
20	80
Patient's age in years	Blood pressure in mm/Hg
Specific Gravity	Albumin
1.01	0
Urine specific gravity (1.005-1.025)	Albumin level (0-6)
Blood Glucose	Blood Urea
100	40
Blood glucose level in mg/dL	Blood urea level in mg/dL
Serum Creatinine	Sodium
1.03	120
Serum creatinine level in mg/dL	Sodium level in mEq/L
Potassium	Hemoglobin
3.5	12
Potassium level in mEq/L	Hemoglobin level in g/dL

Predict

Figures 5, 6 : Prediction Page Before and After Values Entering.

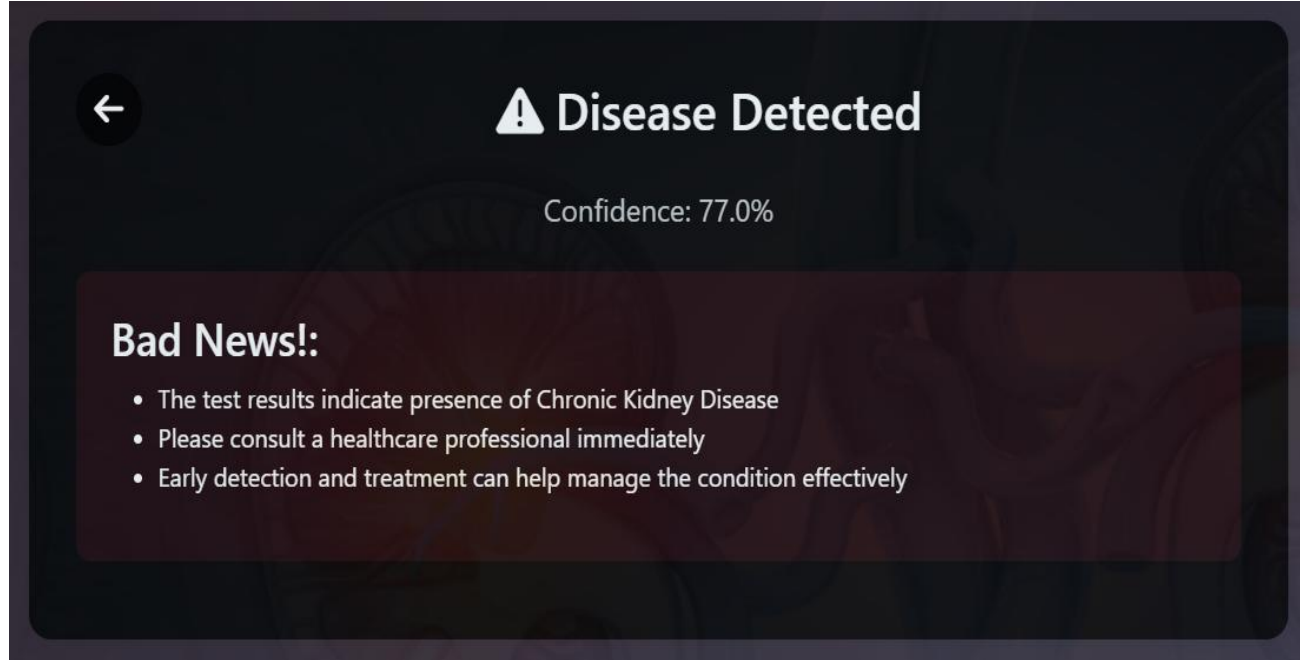


Figure 7: Predicted Result for Disease Detected with 77.0% Accuracy.

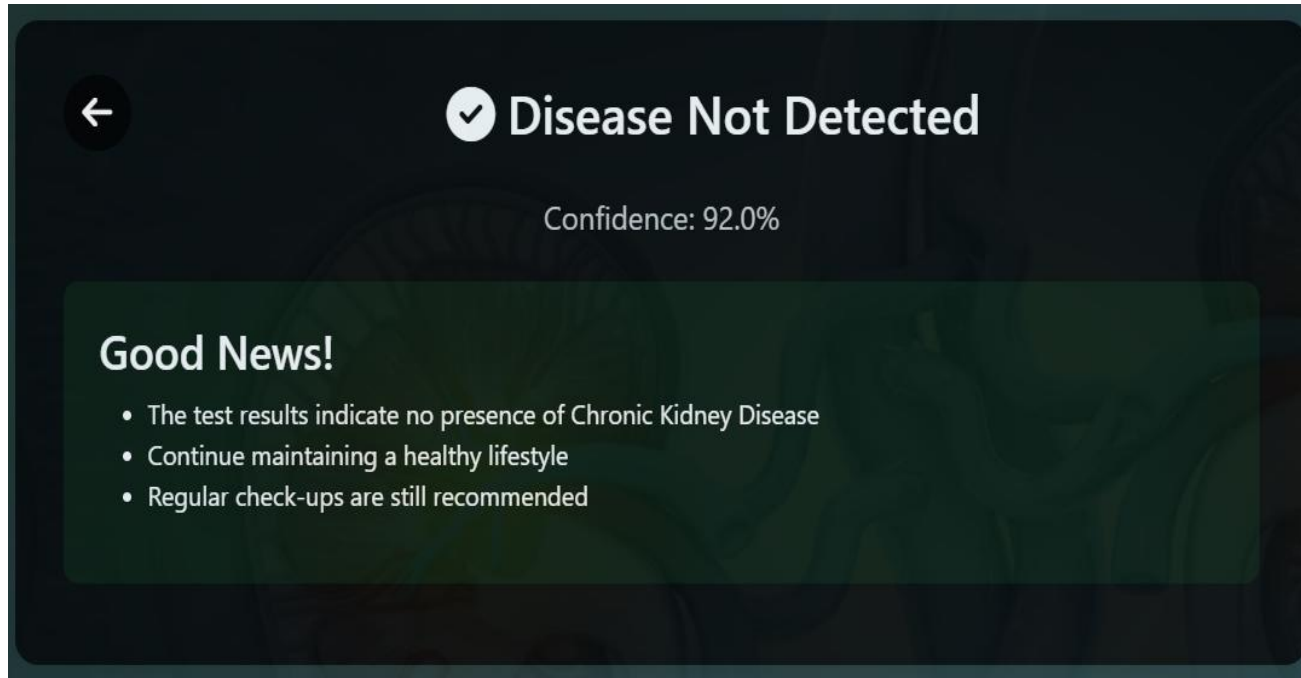


Figure 8 : Predicted Result for Disease not Detected with 92.0% Accuracy.

Comparative Study

- Machine learning is transforming kidney disease prediction, offering a faster and more automated approach to diagnosis. Various models like Logistic Regression, Decision Trees, Random Forest, SVM, KNN, ANN, and XGBoost are used for this purpose. XGBoost and ANN provide the highest accuracy, while Random Forest balances accuracy and interpretability, making it ideal for real-world applications.

- Logistic Regression and Decision Trees are preferred for their transparency in decision-making but may struggle with complex patterns. The best model depends on the need for accuracy, interpretability, or efficiency.
- Future advancements like hybrid models, deep learning (CNNs), and federated learning can enhance kidney disease detection while ensuring patient privacy. Machine learning enables diagnosis and better healthcare outcomes, supporting doctors in making faster, more accurate decisions.

Paper Published



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CERTIFICATE



This is to Certify that **Mr. Yarlanki kirankumar**
of **GODAVARI INSTITUTE OF ENGINEERING AND TECHNOLOGY**
has successfully Presented the Paper Titled
KIDNEY DISEASE PREDICTION



at the 5th International Conference on Navigating Complexity : Harnessing
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Conclusion

This study demonstrates that Logistic Regression is a reliable and interpretable model for predicting kidney disease based on medical parameters. With an accuracy of 89%, the model effectively identifies individuals at risk, enabling timely interventions.

Future work will focus on:

- Incorporating real-time patient data from wearable devices.
- Expanding the dataset to include diverse demographics.

- Exploring advanced techniques like ensemble methods for improved accuracy.

By leveraging machine learning, this study contributes to the growing field of AI in healthcare, offering a scalable and cost-effective solution for kidney disease prediction.

*Thank
you*

